

# A Tower-Recursive Source for Every Gold Diagonal

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## Abstract

Let  $m$  be a power of two, let  $K_m = \mathbb{F}_{2^m}$  be the canonical finite nim-field, and write  $e_i = 2^i$  for its Conway bit basis. For the Gold form

$$Q_a(x) = \text{Tr}_{K_m/\mathbb{F}_2}(x^{1+2^a}),$$

the diagonal constants  $Q_a(e_i)$  are the datum missing from the polar form. Previously a uniform game-built source was known only for  $a = 1$ , while the trace-dual elements for even  $a$  visibly drifted under tower doubling. We give a recursion for the trace-dual diagonal of every scaled Gold component, prove that the unscaled dual always descends to the half-field, and conclude that it is an Artin–Schreier value  $w^2 + w$ . Thus every Gold exponent has a single tower-constructed source  $w$  satisfying

$$\text{Tr}((w^2 + w)e_i) = Q_a(e_i) \quad (0 \leq i < m).$$

The result closes the diagonal-sourcing problem, including the even- $a$  case. It does not solve the separate problem of turning those constants into a non-clock normal-, misère-, or loopy-outcome game.

## 1 Statement and boundary

For a scale  $c \in K_m$ , define the scaled Gold component

$$Q_{a,c}^{(m)}(x) = \text{Tr}_m(cx \text{Fr}^a(x)), \quad \text{Fr}(x) = x^2,$$

where  $\text{Tr}_m = \text{Tr}_{K_m/\mathbb{F}_2}$ . Its canonical-basis diagonal extends uniquely to an  $\mathbb{F}_2$ -linear functional

$$\ell_{a,c}^{(m)}\left(\sum_i x_i e_i\right) = \sum_i x_i Q_{a,c}^{(m)}(e_i).$$

The absolute trace pairing on a finite separable field is nondegenerate, so there is a unique  $\Lambda_{a,c}^{(m)} \in K_m$  such that

$$\text{Tr}_m(\Lambda_{a,c}^{(m)} e_i) = \text{Tr}_m(c e_i \text{Fr}^a(e_i)) \quad \text{for every } i < m. \quad (1)$$

Write  $\Lambda_a^{(m)} = \Lambda_{a,1}^{(m)}$ .

**Theorem 1.1** (all-exponent Gold-diagonal source). *For every power of two  $m \geq 2$  and every  $a \geq 0$ :*

1.  $\Lambda_{a,c}^{(m)}$  is given by the quadratic-tower recursion in Theorem 2.2, using only nim addition and multiplication, Frobenius, relative trace, and the canonical tower generator;

2. the unscaled dual descends:

$$\Lambda_a^{(m)} \in K_{m/2} \subset K_m;$$

3. there is  $w_a^{(m)} \in K_m$  with

$$\Lambda_a^{(m)} = \wp(w_a^{(m)}) = (w_a^{(m)})^2 + w_a^{(m)};$$

and therefore

$$Q_a(e_i) = \text{Tr}_m\left(\wp(w_a^{(m)})e_i\right) \quad (0 \leq i < m).$$

The two possible sources are  $w_a^{(m)}$  and  $w_a^{(m)} + 1$ .

The theorem is about sourcing the diagonal constants. It does not infer a  $P$ -set theorem, a linking strategy, or a resolution of the full natural Gold-quadric game problem. This proof/semantics boundary is essential.

## 2 The quadratic nim-tower block

Put  $m = 2M$ . The canonical nim tower has

$$K_m = K_M(u) = K_M \oplus uK_M, \quad u = e_M,$$

and the nontrivial  $K_M$ -automorphism  $\sigma = \text{Fr}^M$  satisfies

$$\sigma(u) = u + 1.$$

The canonical basis splits as

$$e_0, \dots, e_{M-1}, \quad ue_0, \dots, ue_{M-1};$$

indeed  $ue_j = e_{M+j}$ . Let

$$T = T_{m/M} = 1 + \sigma : K_m \longrightarrow K_M$$

be the relative trace.

**Lemma 2.1** (trace-pairing blocks). *For  $A, B, e \in K_M$ ,*

$$T((A + uB)e) = Be, \tag{2}$$

$$T((A + uB)ue) = (A + B)e. \tag{3}$$

*Proof.* Because  $\sigma$  fixes  $A, B, e$  and sends  $u$  to  $u + 1$ ,

$$(A + uB)e + \sigma((A + uB)e) = (A + uB)e + (A + (u + 1)B)e = Be.$$

For the upper block, apply the same calculation to  $(A + uB)u$ :

$$\begin{aligned} (A + uB)u + (A + (u + 1)B)(u + 1) \\ = Au + u^2B + Au + A + (u + 1)^2B = A + B, \end{aligned}$$

since characteristic two cancels the doubled  $Au$ ,  $u^2B$ , and  $uB$  terms. Multiplication by  $e$  gives (3).  $\square$

Now write  $c = c_0 + uc_1$  and set

$$d_0 = T(c), \quad d_1 = T(cu \operatorname{Fr}^a(u)). \quad (4)$$

For a lower basis vector  $e \in K_M$ ,

$$Q_{a,c}^{(m)}(e) = \operatorname{Tr}_M(T(c)e \operatorname{Fr}^a(e)) = Q_{a,d_0}^{(M)}(e).$$

For its upper mate  $ue$ ,

$$Q_{a,c}^{(m)}(ue) = \operatorname{Tr}_M(T(cu \operatorname{Fr}^a(u))e \operatorname{Fr}^a(e)) = Q_{a,d_1}^{(M)}(e).$$

Thus  $d_0$  and  $d_1$  are exactly the smaller-field scales of the two diagonal blocks.

**Theorem 2.2** (scaled dual recursion). *Let*

$$L_0 = \Lambda_{a,d_0}^{(M)}, \quad L_1 = \Lambda_{a,d_1}^{(M)},$$

with  $d_0, d_1$  as in (4). Then

$$\Lambda_{a,c}^{(2M)} = (L_0 + L_1) + uL_0. \quad (5)$$

The base case over  $K_1 = \mathbb{F}_2$  is  $\Lambda_{a,c}^{(1)} = c$ .

*Proof.* Write the candidate in (5) as  $A + uB$ , where  $B = L_0$  and  $A = L_0 + L_1$ . By Lemma 2.1, its relative trace pairing with a lower basis vector  $e$  is  $L_0e$ , while its pairing with  $ue$  is

$$(A + B)e = (L_0 + L_1 + L_0)e = L_1e.$$

After applying  $\operatorname{Tr}_M$ , these are exactly the two defining dualities for the lower and upper diagonal blocks. Nondegeneracy of the absolute trace pairing makes the candidate unique, proving (5).  $\square$

This is the point at which the even- $a$  drift ceases to be an obstruction. The scale  $d_1$  can change at every tower level; the recursion transports that change instead of requiring one fixed dual element throughout the tower.

### 3 Descent and the Artin–Schreier source

**Lemma 3.1** (unscaled half-field descent). *For  $m = 2M$ ,*

$$\Lambda_a^{(m)} \in K_M.$$

*Proof.* In the unscaled case  $c = 1$ , the lower scale is

$$d_0 = T(1) = 1 + 1 = 0.$$

Linearity (or the recursion from the base case) gives  $L_0 = \Lambda_{a,0}^{(M)} = 0$ . Theorem 2.2 therefore reduces to

$$\Lambda_a^{(2M)} = L_1 \in K_M.$$

$\square$

**Lemma 3.2** (Artin–Schreier exact sequence). *For a finite field  $K = \mathbb{F}_{2^m}$ ,*

$$\text{im}(x \mapsto x^2 + x) = \ker(\text{Tr}_{K/\mathbb{F}_2}).$$

*Each element of this common hyperplane has exactly two preimages, differing by 1.*

*Proof.* The Artin–Schreier map  $\wp(x) = x^2 + x$  is  $\mathbb{F}_2$ -linear and has kernel  $\mathbb{F}_2 = \{0, 1\}$ , so its image has  $|K|/2$  elements. Frobenius invariance of trace gives

$$\text{Tr}(\wp(x)) = \text{Tr}(x^2) + \text{Tr}(x) = \text{Tr}(x)^2 + \text{Tr}(x) = 0,$$

so the image lies in the trace-zero hyperplane. The finite-field trace is a nonzero  $\mathbb{F}_2$ -linear functional, hence its kernel also has  $|K|/2$  elements. The inclusion is equality. Every nonempty fibre is a coset of the two-element kernel.  $\square$

*Proof of Theorem 1.1.* Theorem 2.2 proves the constructive first assertion and Lemma 3.1 proves the second. If  $\lambda = \Lambda_a^{(2M)} \in K_M$ , trace transitivity gives

$$\text{Tr}_{2M}(\lambda) = \text{Tr}_M(\text{Tr}_{K_{2M}/K_M}(\lambda)) = \text{Tr}_M(2\lambda) = 0.$$

Lemma 3.2 supplies  $w$  with  $w^2 + w = \lambda$ . Substituting this identity into the defining trace duality (1) yields the stated formula for every basis coin. The fibre statement in Lemma 3.2 gives the two sources  $w$  and  $w + 1$ .  $\square$

## 4 Implementation and formal boundary

The Rust core implements the proof literally:

- `gold_component_diagonal_dual(m,a,c)` evaluates Theorem 2.2;
- `gold_diagonal_dual(m,a)` specializes to  $c = 1$ ; and
- `gold_diagonal_artin_schreier_source(m,a)` uses the existing exact finite-nim-field solver to choose one of the two sources.

The implementation is checked directly against all defining trace-pairing equations for every scale over  $K_m$  at  $m \leq 8$ . For the unscaled theorem it checks half-field descent, the Artin–Schreier equation, and every diagonal coordinate at representative exponents through the full  $m = 128$  backend. The previous even- $a$  values

$$\Lambda_2^{(4)} = 0, \quad \Lambda_2^{(8)} = 6, \quad \Lambda_2^{(16)} = 102, \quad \Lambda_2^{(32)} = 24582$$

and  $\Lambda_4^{(16)} = 31$ ,  $\Lambda_4^{(32)} = 8030$  are pinned as regression values; their drift is now explained and handled by the scale recursion.

The Lean module `formal/Ogdoad/GoldDiagonal.lean` kernel-checks the quadratic trace-block formulas, the dual-block reconstruction, the vanishing of the absolute trace of a quadratic-extension base element, additivity of the Artin–Schreier map, the one-step tower lift of a source, and the full finite-field equivalence  $\text{Tr}(\lambda) = 0 \iff \lambda = w^2 + w$ . The last result is proved from rank-nullity, trace surjectivity, and the two-element kernel, rather than introduced as an axiom. The module does not encode the finite nim tower or replace the Rust implementation with a Lean finite-field model.

## 5 Consequence for the open Gold-game program

The former even- $a$  constructive sourcing question is closed: for every exponent, the diagonal is supplied by one deterministically selected tower constant through nim multiplication, Frobenius, trace, and finally one square plus XOR at the source identity  $\Lambda = \wp(w)$ . No coordinate-by-coordinate evaluation of  $Q_a$  occurs in the recursion. Unlike the  $a = 1$  Fermat-coin formula, the final  $w$  is selected by exact finite-field linear algebra; the theorem does not claim a comparably simple closed expression or that this selection itself satisfies the strategic naturality axioms.

What remains open is the strategic layer. A rule may consume the sourced diagonal and still be a transport clock, or may realize only a terminal charge rather than a normal/misère/loopy outcome class. The all- $a$  source removes one algebraic obstruction from that problem; it does not weaken the existing naturality and outcome-semantics requirements.